The editing analogy

A nurse opens a patient record and starts editing. She changes the phone number. Corrects the address. Updates the GP. She hasn't hit Save yet. **Where are those changes living right now?**

Working in RAM	Committing to disk
Happens while you edit	Happens when you hit Save
Fast and cheap to change	Permanent and protected
Change your mind freely	Transaction confirmed ✓
No disk writes yet	Data written to disk

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The same principle applies to your SQL queries

When you run a query, the database does all its work in memory.

Disk	→	**RAM**	→	**You**
(data lives here)	→	(database works here)	→	(results returned)
		GROUP BY happens		
		SUM calculated		
		ORDER BY sorted		

Key points:

- Data is pulled off disk into RAM
- All aggregation, filtering, joining, sorting happens in RAM
- Only the finished result comes back to you
- Writing to disk at every step would be impossibly slow

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Why RAM is faster (the physics)

Headline: The speed difference isn't engineering — it's physics.

Three-row comparison:

Storage type	How it works	Speed
RAM	Electrical charges in transistors — no moving parts	Nanoseconds
SSD	Flash memory chips — no moving parts but slower interface	Microseconds
Hard Drive (HDD)	Spinning magnetic disk + physical arm that moves	Milliseconds

The scaled comparison to show prominently:

If RAM = **1 second** SSD = **4 minutes** Hard drive = **2 hours** Same data. Different location.

The warning to include:

If a query is too large to fit in RAM, the database spills to disk temporarily — and that is when queries get very slow.

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The implication for your SQL

Precise query → pulls only what it needs → fast

Sloppy query → pulls everything → forces unnecessary disk reads → slow

Example:

```
-- Sloppy — pulls every column, every row
SELECT *
FROM Appointments

-- Precise — pulls only what the query needs
SELECT ConsultantID, AppointmentDate
FROM Appointments
WHERE AppointmentDate >= '2024-01-01'
```

Key message:

Every extra column and every extra row you pull forces more data off disk into memory.

Write precisely — only ask for what you need.

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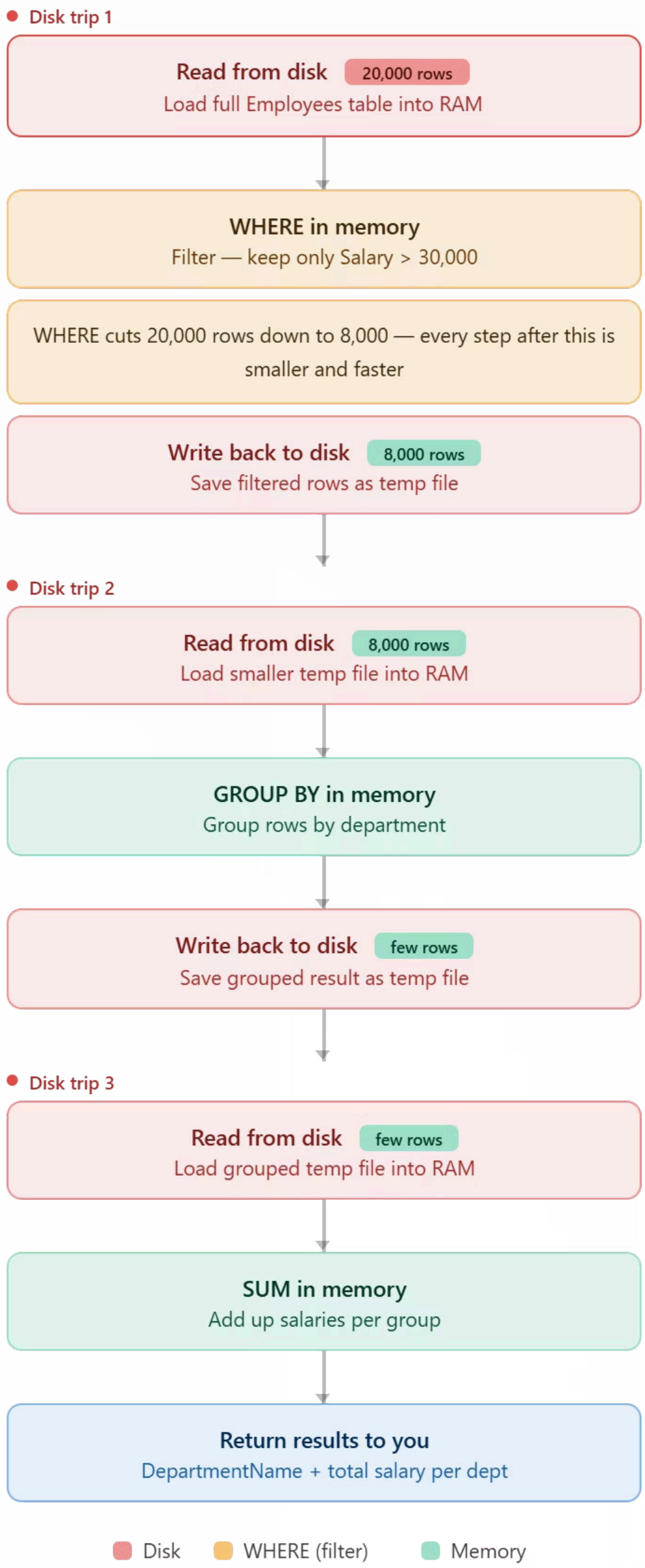
```
SELECT DepartmentName, SUM(Salary)
FROM Employees
WHERE Salary > 30000
GROUP BY DepartmentName
```



Same **3 disk trips** as before — but after WHERE, every trip carries **far less data**.

WHERE doesn't remove trips — it shrinks everything that follows.

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QUIZ

What would you do if you had a large data set on disk and only a small amount of RAM to work with and you needed to sort the data?

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Glassdoor

Google Interview Question: What sort would you use if you had a large data set on disk and a small amount of ram to wor...

Interview question for Senior Software Engineer.What sort would you use if you had a large data set on disk and a small amount of ram to work with?.

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Why Spark is faster — it's all about memory

The same principle: read from disk once → work in memory → write once. Three tools, three approaches.

Why Spark is faster — it's all about memory

The same principle: read from disk once → work in memory → write once. Three tools, three approaches.

TRADITIONAL SQL

Keeps running back to disk between every step

Read from disk
↓ work ↓
Write back to disk
↓ repeat ↓
Read from disk again
↓ work ↓
Write back to disk

Fine at small scale. At billions of rows, all those round trips make it painfully slow.

PYTHON / PANDAS

Loads everything into one machine's RAM — then crashes if it doesn't fit

Read ALL from disk
↓
Work in RAM
↓
Work in RAM
↓
Work in RAM
↓
Done

Fast — until data exceeds RAM. Then it has no plan B. It just crashes.

APACHE SPARK

Reads once, chains everything in memory, writes once — across many machines

Read from disk — once
↓
Filter in RAM
↓
Join in RAM
↓
Aggregate in RAM
↓
Write result — once

Distributed across many machines in parallel. Data too big for one machine? No problem.

SQL chef — finishes chopping, walks back to the fridge, puts ingredients away, comes back out. Every. Single. Step.

Pandas chef — brings everything out of the fridge at once. Fast — until the counter is too small and everything spills.

Spark kitchen — a whole team of chefs, each with their own counter. Ingredients come out once. Nobody goes back to the fridge until the dish is done.

The rule: every unnecessary trip to disk costs time. Spark wins because it makes that trip **once** — and shares the work across an entire team.

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Students

StudentID PK	FirstName	LastName	DateOfBirth	Email
S001	Amara	Jones	12 Mar 2000	a.jones@student.ac.uk
S002	James	Wright	05 Jul 1999	j.wright@student.ac.uk
S003	Priya	Sharma	22 Nov 2001	p.sharma@student.ac.uk
S004	Kwame	Asante	18 Feb 2000	k.asante@student.ac.uk
S005	Chloe	Martin	30 Aug 1998	c.martin@student.ac.uk

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Teachers

TeacherID PK	FirstName	LastName	Email	Department
T01	Ravi	Patel	r.patel@school.ac.uk	Data & Computing
T02	Blessing	Okafor	b.okafor@school.ac.uk	Data & Computing
T03	Wei	Chen	w.chen@school.ac.uk	Cloud & DevOps
T04	Ama	Mensah	a.mensah@school.ac.uk	AI & Analytics

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Courses

CourseID PK	CourseName	Credits	TeacherID FK
C01	SQL Basics	10	**T01**
C02	Python Intro	10	**T02**
C03	Data Modelling	15	**T01**
C04	Cloud Basics	10	**T03**
C05	AI Fundamentals	15	**T04**

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Enrolments

EnrolmentID PK	StudentID FK	CourseID FK	EnrolDate	Grade
E001	**S001**	**C01**	05 Sep 2024	A
E002	**S001**	**C02**	05 Sep 2024	B
E003	**S002**	**C01**	05 Sep 2024	C
E004	**S003**	**C03**	06 Sep 2024	A
E005	**S004**	**C05**	06 Sep 2024	B
E006	**S005**	**C04**	07 Sep 2024	A
E007	**S002**	**C03**	07 Sep 2024	B
E008	**S004**	**C02**	08 Sep 2024	A

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